

# HAOS-IIP HBP Recovery Benchmarks: Harder Temporal Recovery and Cross-Domain Transfer Under Frozen Negative Results

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June 2026

## Abstract

This release records the current HBP recovery-benchmark branch as a numbered paper companion. It is not a new phase, not a new mechanism, and not a physical bridge claim. Its function is to make the benchmark harder to win in places where ordinary graph methods already dominate, while preserving a frozen negative result whenever the evidence does not support a stronger claim.

The main benchmarks are PB-03, Temporal Recovery Under Damage, and PB-04, Cross-Domain Structure Transfer. PB-03 asks whether HAOS-style recovery descriptors can predict future recoverability of a damaged system better than topology-only methods on untouched holdout. PB-04 asks whether the same recoverability structure can transfer across domains without learning domain-specific shortcuts. Both benchmarks are frozen, deterministic, and still negative: after hardening the feature layer and the holdout split discipline, PB-03 remains `PREDICTION_NOT_DISTINCT_FROM_BASELINES` with result hash `pb03_result_c989483149438ed29eabc86d`; PB-04 remains `PREDICTION_NOT_DISTINCT_FROM_BASELINES` with result hash `pb04_result_7b13bc9285cc288cbf8e46b8`. That outcome is the intended one for this paper: the benchmarks are getting stricter, not prettier.

## 1 Status

This paper is release 67.1 in the HAOS-IIP numbered paper spine. It is a release companion for the HBP registry and the PB-01 through PB-04 benchmark arc.

It records:

- the frozen HBP bridge registry;
- the PB-01 network recovery benchmark and its negative control discipline;
- the harder PB-03 temporal recovery benchmark;
- the harder PB-04 cross-domain transfer benchmark;
- the current negative results and their deterministic hashes;
- the explicit boundary that a negative result is not a failure of the instrument.

It is not:

- a new phase activation;
- a new physical claim;
- a claim that HAOS-IIP predicts recovery better than graph baselines;
- a claim that cross-domain transfer has succeeded;
- a claim that the current benchmarks are final.

## 2 Why This Release Exists

The HBP program exists to ask a narrower question than the larger physics-facing branches:

Can HAOS-IIP produce a recovery benchmark where a future win would mean something rather than merely matching what topology already gives away?

The answer needs to be auditable in both directions. If HAOS wins too easily, the benchmark is too weak. If HAOS loses honestly, the benchmark is doing its job. The present release therefore emphasizes hardening over celebration.

## 3 Registry Snapshot

The HBP registry now contains the bridge program scaffolding, benchmark contracts, tests, and frozen result artifacts. The key constraint is that claim strength is bounded by the evidence already observed.

| Bridge | Role                              | Current status  |
|--------|-----------------------------------|-----------------|
| PB-01  | Network recovery bridge           | frozen negative |
| PB-02  | External network recovery holdout | frozen negative |
| PB-03  | Temporal recovery under damage    | harder negative |
| PB-04  | Cross-domain structure transfer   | harder negative |

## 4 PB-03: Temporal Recovery Under Damage

PB-03 asks whether HAOS can predict future recoverability of a damaged system better than topology-only methods.

The benchmark is intentionally harder than a static graph regression:

- observe a network at  $t_0$ ;
- introduce damage at  $t_1$ ;
- predict recovery potential at  $t_2$ ;
- compare against degree, betweenness, shortest-path, spectral, and supervised graph baselines;
- preserve a frozen development/calibration/holdout split;
- keep the verdict on untouched holdout.

The current implementation hardens the descriptor layer around recovery-oriented temporal signatures rather than raw ordering alone. The split is also frozen by case identity hash rather than by array order, reducing accidental leakage from source-file layout.

The current result remains negative:

PREDICTION\_NOT\_DISTINCT\_FROM\_BASELINES

with result hash

pb03\_result\_c989483149438ed29eabc86d.

That is a bounded outcome, not a disappointment. It means the benchmark still resists a cosmetic win.

## 5 PB-04: Cross-Domain Structure Transfer

PB-04 asks whether a recoverability signature can transfer across domains without learning domain-specific shortcuts.

The target discipline is stricter than ordinary graph transfer:

- the source and target domains are frozen;
- the transfer protocol is frozen;
- the evaluation is frozen;
- the source features are reduced to a shared recoverability vocabulary;
- the target features are built from the same kind of descriptor family rather than a domain-specific topology cheat;
- the verdict is determined on the frozen target bundle.

The current run stays negative:

PREDICTION\_NOT\_DISTINCT\_FROM\_BASELINES

with result hash

pb04\_result\_7b13bc9285cc288cbfbe46b8.

This is the useful outcome because it says the instrument still has not discovered a transfer shortcut by accident. The benchmark remains open for a genuinely stronger future result.

## 6 What Changed in the Benchmark

Two small but meaningful hardening steps were applied before this release:

- PB-03 now splits by case identity hash rather than raw selected order.
- PB-04 now uses a stricter shared recoverability descriptor vocabulary instead of leaning on easy graph-like summary cues.

These are benchmark changes, not mechanism changes. They reduce the chance that a generic graph regressor wins for the wrong reason.

## 7 What This Release Does Not Claim

This paper does not claim:

- that HAOS-IIP explains network recovery;
- that HAOS-IIP currently beats graph baselines on holdout;
- that recoverability transfer is established;
- that any negative result should be turned into a positive by tuning;
- that the benchmarks are already maximal or final.

The paper exists because a good negative result is not a dead end. It is a better measurement boundary.

## 8 Public Release Data

| Artifact          | Value                                                          | Meaning                                                   |
|-------------------|----------------------------------------------------------------|-----------------------------------------------------------|
| PB-03 result hash | <code>pb03_result_c989483149486cd291b186d48cd291b186d</code>   | negative holdout result                                   |
| PB-04 result hash | <code>pb04_result_7b13bc9285cd288c1f1e46b8d288c1f1e46b8</code> | negative transfer result                                  |
| HBP suite         | 28 tests                                                       | frozen bundle still passes                                |
| Current verdict   | negative                                                       | benchmarks remain hard enough to refuse a superficial win |

## 9 Conclusion

The HBP recovery program is where the architecture now has to earn any future positive claim. PB-03 asks whether future recoverability can be predicted. PB-04 asks whether recoverability transfers across domains. Both are now harder, more isolated from easy graph shortcuts, and still negative.

That is a good state to publish.